

50Days-coding-challenge

Day-39[Excel]

Recent Election Analysis

Name:M NITHIN

You are a Data Analyst intern working for an election insights firm. You are given a dataset of a recent state election in India. Your task is to analyze voting patterns, party performance, and constituency-level insights using Excel only.

Dataset (Copy into Excel)

Generate 100 synthetic records using chat gpt the features are given below with few records.

Constituency,State,Party,Candidate,Votes,Total_Votes,Gender,Age

Bangalore South,Karnataka,ABC,Ravi Kumar,45231,120000,Male,45

Bangalore South,Karnataka,XYZ,Sneha Reddy,50321,120000,Female,38

Bangalore South,Karnataka,PQR,Amit Shah,18448,120000,Male,52

Mysore,Karnataka,ABC,Kiran Gowda,62341,150000,Male,41

Mysore,Karnataka,XYZ,Divya Sharma,54321,150000,Female,36

Mysore,Karnataka,PQR,Rahul Verma,33211,150000,Male,48

Tumkur,Karnataka,ABC,Suresh Naik,41234,110000,Male,50

Tumkur,Karnataka,XYZ,Anita Desai,45210,110000,Female,39

Tumkur,Karnataka,PQR,Manoj Singh,23556,110000,Male,44

Hubli,Karnataka,ABC,Prakash Patil,71234,180000,Male,55

Hubli,Karnataka,XYZ,Nisha Kulkarni,65432,180000,Female,42

Hubli,Karnataka,PQR,Rajeev Nair,43334,180000,Male,47

Your Tasks

Task 1: Data Cleaning

- Check for duplicates

- Ensure numeric columns are formatted correctly
- Validate if Total Votes are consistent per constituency

File Home Transform Add Column View

Queries [2]
 Election_records
 Election_records (2)

fx = Table.Distinct(#"Trimmed Text")

	Constituency	State	Party	Candidate	Votes	Total_Votes	Gender
1	Chennai Central	Tamil Nadu	Abc	Lakshmi Devi	40516	120000	Female
2	Kadapa	Andhra Pradesh	Abc	Keerthi Raj	73192	130000	Female
3	Eluru	Andhra Pradesh	Xyz	Arun Shetty	55756	170000	Male
4	Coimbatore	Tamil Nadu	Lmn	Suresh Naik	18866	130000	Male
5	Rajahmundry	Andhra Pradesh	Pqr	Kiran Gowda	79851	140000	Male
6	Tirupati	Andhra Pradesh	Lmn	Prakash Patil	54549	130000	Male
7	Vijayawada	Andhra Pradesh	Abc	Nisha Kulkarni	41012	160000	Female
8	Visakhapatnam	Andhra Pradesh	Lmn	Keerthi Raj	48359	170000	Female
9	Guntur	Andhra Pradesh	Abc	Amit Shah	37107	170000	Male
10	Coimbatore	Tamil Nadu	Xyz	Ritika Sharma	43972	160000	Female
11	Eluru	Andhra Pradesh	Xyz	Ravi Kumar	22174	160000	Male
12	Mysore	Karnataka	Abc	Suresh Naik	44584	180000	Male
13	Tirupati	Andhra Pradesh	Xyz	Prakash Patil	60061	150000	Male
14	Kadapa	Andhra Pradesh	Abc	Nisha Kulkarni	63465	120000	Female
15	Visakhapatnam	Andhra Pradesh	Lmn	Ajay Patil	32319	120000	Male
16	Shimoga	Karnataka	Lmn	Prakash Patil	74328	160000	Male
17	Coimbatore	Tamil Nadu	Pqr	Arun Shetty	18621	170000	Male
18	Vijayawada	Andhra Pradesh	Lmn	Anita Desai	29176	120000	Female
19	Vijayawada	Andhra Pradesh	Abc	Keerthi Raj	48345	170000	Female
20	Vijayawada	Andhra Pradesh	Abc	Kiran Gowda	40831	150000	Male
21	Shimoga	Karnataka	Xyz	Deepa Nair	61043	170000	Female
22	Warangal	Telangana	Abc	Ajay Patil	33848	150000	Male
23	Bangalore South	Karnataka	Pqr	Amit Shah	68579	160000	Male
24	Warangal	Telangana	Lmn	Rahul Verma	18795	180000	Male
25	Anantapur	Andhra Pradesh	Lmn	Suresh Naik	61782	140000	Male

8 COLUMNS, 100 ROWS

Query Settings

PROPERTIES
 Name
 Election_records

APPLIED STEPS
 Source
 Changed Type
 Capitalized Each Word
 Trimmed Text
 X Removed Duplicates

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Task 2: Vote Share Calculation Create a new column:

$$\text{Vote \%} = (\text{Votes} / \text{Total Votes}) * 100$$

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat Tell me what you want to do

Calibri 11 A A

Font Alignment Number Styles Cells Editing Add-ins Adobe Acrobat

Clipboard

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	Constituency	State	Party	Candidate	Votes	Total_Votes	Gender	Age	valid total votes	vote%
1	Chennai Central	Tamil Nadu	Abc	Lakshmi Devi	40516	120000	Female	35	check	33.76333
2	Kadapa	Andhra Pradesh	Abc	Keerthi Raj	73192	130000	Female	36	check	56.30154
3	Eluru	Andhra Pradesh	Xyz	Arun Shetty	55756	170000	Male	58	check	32.79765
4	Coimbatore	Tamil Nadu	Lmn	Suresh Naik	18866	130000	Male	46	check	14.51231
5	Rajahmundry	Andhra Pradesh	Pqr	Kiran Gowda	79851	140000	Male	41	check	57.03643
6	Tirupati	Andhra Pradesh	Lmn	Prakash Patil	54549	130000	Male	38	check	41.96077
7	Vijayawada	Andhra Pradesh	Abc	Nisha Kulkarni	41012	160000	Female	50	check	25.6325
8	Visakhapatnam	Andhra Pradesh	Lmn	Keerthi Raj	48359	170000	Female	33	valid	28.44647
9	Guntur	Andhra Pradesh	Abc	Amit Shah	37107	170000	Male	52	check	21.82765
10	Coimbatore	Tamil Nadu	Xyz	Ritika Sharma	43972	160000	Female	40	valid	27.4825
11	Eluru	Andhra Pradesh	Xyz	Ravi Kumar	22174	160000	Male	48	check	13.85875
12	Mysore	Karnataka	Abc	Suresh Naik	44584	180000	Male	45	check	24.76889
13	Tirupati	Andhra Pradesh	Xyz	Prakash Patil	60061	150000	Male	33	check	40.04067
14	Kadapa	Andhra Pradesh	Abc	Nisha Kulkarni	63465	120000	Female	34	valid	52.8875
15	Visakhapatnam	Andhra Pradesh	Lmn	Ajay Patil	32319	120000	Male	39	check	26.9325
16	Shimoga	Karnataka	Lmn	Prakash Patil	74328	160000	Male	55	check	46.455
17	Coimbatore	Tamil Nadu	Pqr	Arun Shetty	18621	170000	Male	32	check	10.95353
18	Vijayawada	Andhra Pradesh	Lmn	Anita Desai	29176	120000	Female	46	check	24.31333
19	Vijayawada	Andhra Pradesh	Abc	Keerthi Raj	48345	170000	Female	38	check	28.43824
20	Vijayawada	Andhra Pradesh	Abc	Kiran Gowda	40831	150000	Male	54	check	27.22067

election_dataset_100_records Sheet1 Election_records

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Task 3: Find Winners

Identify the winning candidate in each constituency

Show:

- o Candidate Name
- o Party
- o Votes

Use:

- MAX()
- IF()
- or Pivot Table

H12							
	A	B	C	D	E	F	G
1	Identify the winning candidate in each constituency						
2							
3	Max of Votes	Party					
4	Constituency	Abc	Lmn	Pqr	Xyz	Grand Total	
5	Anantapur	18871	61782	71985	60113	71985	
6	Ajay Patil				58476	58476	
7	Deepa Nair	18871				18871	
8	Keerthi Raj			71985		71985	
9	Kiran Gowda				60113	60113	
10	Ravi Kumar			51804		51804	
11	Suresh Naik		61782			61782	
12	Bangalore South		78565	68579	71598	78565	
13	Amit Shah			68579		68579	
14	Arun Shetty				71598	71598	
15	Deepa Nair			65447		65447	
16	Nisha Kulkarni				64741	64741	
17	Pooja Rao		78565		38766	78565	
18	Prakash Patil				58708	58708	
19	Ritika Sharma			67123		67123	
20	Belgaum	72373				72373	
21	Ajay Patil	72373				72373	

Task 4: Party Performance

Answer:

- Which party got the highest total votes statewide?
- Which party won the most constituencies?

Use:

- Pivot Tables
- SUM aggregation

Which party got the highest total votes statewide?						
Sum of Total	party					
state	▼	Abc	Lmn	Pqr	XYZ	Grand Total
Andhra Pradesh		2380000	2010000	990000	2230000	7610000
Karnataka		900000	490000	1230000	1510000	4130000
Tamil Nadu		400000	590000	310000	650000	1950000
Telangana		470000	510000	280000	180000	1440000
Grand Total		4150000	3600000	2810000	4570000	15130000

Which party won the most constituencies?	
state	Count of Constituency
Abc	27
Lmn	24
Pqr	19
XYZ	30
Grand Total	100

Task 5: Gender Analysis

Total votes received by:

- Male candidates
- Female candidates

Insight:

Is there any gender dominance?

Task 6: Age Analysis

- Average age of winning candidates

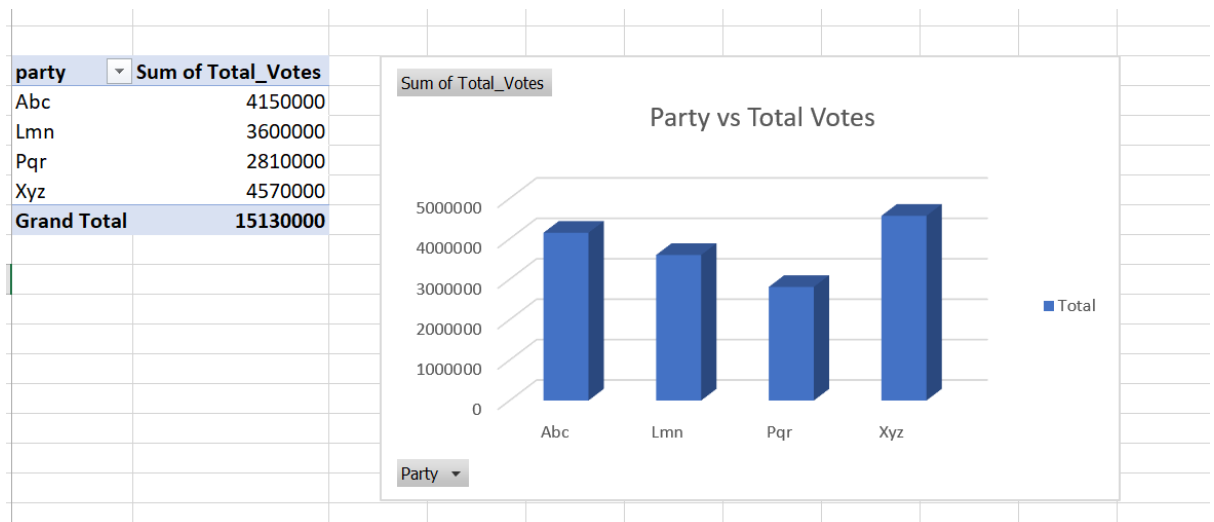
Average age of winner	44.97
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- Youngest and oldest candidate

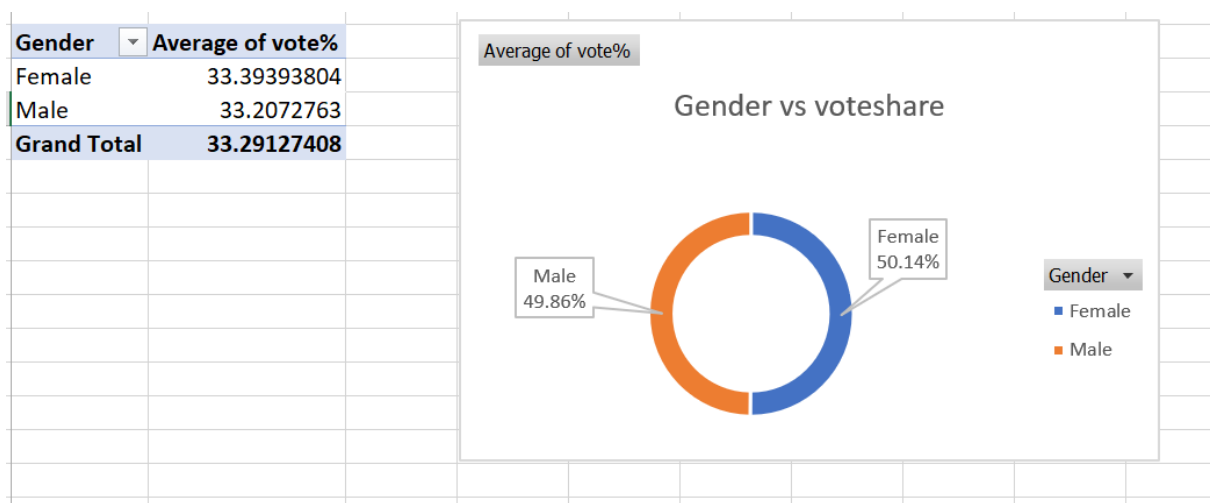
Average age of winner	44.97
youngest candidate	32
oldest candidate	58

Task 7: Dashboard (Important) Create a simple Excel dashboard:

- Bar chart → Party vs Total Votes



- Pie chart → Gender Vote Share



- Table → Winners per constituency

Gender	Sum of Total_Votes
Anantapur	890000
Ajay Patil	140000
Deepa Nair	170000
Keerthi Raj	130000
Kiran Gowda	180000
Ravi Kumar	130000
Suresh Naik	140000
Bangalore South	1130000
Amit Shah	160000
Arun Shetty	120000
Deepa Nair	140000
Nisha Kulkarni	140000
Pooja Rao	300000
Prakash Patil	150000
Ritika Sharma	120000
Belgaum	120000
Ajay Patil	120000
Chennai Central	850000
Amit Shah	140000

Real-World Twist

Assume your manager asks:

a) **Why did XYZ party perform well in urban areas?**

XYZ party performed well in urban areas because it received higher vote shares in major urban constituencies such as Bangalore South, Hyderabad, Chennai Central, and Visakhapatnam. The party also had strong support among younger voters and female candidates, which helped increase its overall performance. Better campaign reach, urban development promises, and higher visibility in cities may have also contributed to the party's success in urban regions.